

### 5.3.5 Embedded voltage reference

The parameters provided in Table 20. Embedded internal voltage reference are derived from tests performed under the ambient temperature and supply voltage conditions summarized in Section 5.3.1.

**Table 20. Embedded internal voltage reference**

The values in this table are specified by design and not tested in production.

| Symbol                                    | Parameter                                                           | Conditions                       | Min   | Typ   | Max   | Unit   |
|-------------------------------------------|---------------------------------------------------------------------|----------------------------------|-------|-------|-------|--------|
| V <sub>REFINT</sub>                       | Internal reference voltages                                         | -40 °C < T <sub>J</sub> < 140 °C | 1.180 | 1.217 | 1.250 | V      |
| t <sub>S_vrefint</sub> <sup>(1)(2)</sup>  | ADC sampling time when reading the internal reference voltage       | -                                | 4.3   | -     | -     | μs     |
| t <sub>S_vbat</sub> <sup>(1)</sup>        | VBAT sampling time when reading the internal VBAT reference voltage | -                                | 9     | -     | -     | μs     |
| t <sub>start_vrefint</sub> <sup>(2)</sup> | Start time of reference voltage buffer when ADC is enable           | -                                | -     | -     | 4.4   | μs     |
| I <sub>refbuf</sub> <sup>(2)</sup>        | Reference Buffer consumption for ADC                                | V <sub>DDA</sub> = 3.3 V         | 9     | 13.5  | 23    | μA     |
| ΔV <sub>REFINT</sub> <sup>(2)</sup>       | Internal reference voltage spread over the temperature range        | -40 °C < T <sub>J</sub> < 140 °C | -     | 5     | 15    | mV     |
| T <sub>coeff</sub>                        | Average temperature coefficient                                     | Average temperature coefficient  | -     | 19    | 67    | ppm/°C |
| V <sub>DDcoeff</sub>                      | Average Voltage coefficient                                         | 3.0 V < V <sub>DD</sub> < 3.6 V  | -     | 10    | 1370  | ppm/V  |

1. The shortest sampling time for the application can be determined by multiple iterations.
2. Specified by design - Not tested in production.

**Table 21. Internal reference voltage calibration values**

| Symbol                  | Parameter                                                           | Memory address          |
|-------------------------|---------------------------------------------------------------------|-------------------------|
| V <sub>REFINT_CAL</sub> | Raw data acquired at temperature of 30 °C, V <sub>DDA</sub> = 3.3 V | 0x08FFF810 - 0x08FFF812 |

### 5.3.6 Supply current characteristics

The current consumption is a function of several parameters and factors such as the operating voltage, ambient temperature, I/O pin loading, device software configuration, operating frequencies, I/O pin switching rate, program location in memory and executed binary code.

The current consumption is measured as described in Current consumption measurement.

#### Typical and maximum current consumption

The MCU is placed under the following conditions:

- All I/O pins are in analog input mode.
- All peripherals are disabled except when explicitly mentioned.
- The flash memory access time is adjusted with the minimum wait-state number, depending on the f<sub>HCLK</sub> frequency (refer to the tables *Number of wait states according to CPU clock (HCLK) frequency* available in the product reference manual).
- When the peripherals are enabled, f<sub>PCLK</sub> = f<sub>HCLK</sub>.

The parameters given in the tables below are derived from tests performed under ambient temperature and supply voltage conditions summarized in Section 5.3.1: General operating conditions. If not specified otherwise, typical data are measured with a V<sub>DD</sub> supply of 3.0 V, and maximum data are measured at 3.6 V.